Lesson notes · for you

How attacks work, and how to defend: unit check

Cyber security · Level 3 · Unit check

Time	Materials
20 min	Paper and a pen. No device needed.

Learning goals

- Malware: how it gets in
- Social engineering: hacking people
- Attacks on passwords, and the defences
- Secret messages: encryption basics

Activity: Defender's map

1. Draw (5 min): Draw your phone in the middle of a page. Around it, draw four arrows for the four attacks: malware, social engineering, password attacks and stolen keys.
2. Defend (8 min): On each arrow, write the defences that block it: official store, updates and backups; pause and check another way; long unique passphrases and two-step sign-in; keep codes and PINs secret.
3. Check (4 min): Tick every defence you already have on your own phone and accounts.
4. Plan (3 min): Circle one defence you don't have yet, and write when you'll set it up.

Keep the map: it's a checklist you can show someone at home.

Discussion

- Which attack in this unit is most likely to reach you, and why?
- Which defence protects against more than one attack?
- Who at home would you most like to teach, and what?

Check yourself

- The one defence I'll set up this week, and when.

Try it this week
Pick the defence from this course that you need most (updates, a backup, unique passphrases or two-step sign-in), set it up this week, and teach it to one person at home.

Worksheet

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Name _____ Class _____ Date _____

1 **Which kind of malware pretends to be something useful, like a free game hack, so you install it yourself?**

◯ A worm ◯ A trojan ◯ Ransomware ◯ Spyware

2 **If you think your phone is infected, turning off Wi-Fi and mobile data is a good first step.**

◯ True ◯ False

3 **A caller says: "I'm from the bank's security team. Your account will be frozen in 15 minutes unless you read me the code we just sent." Which tricks is the caller using?**

☐ Authority ☐ Urgency ☐ Fear ☐ Curiosity ☐ A gift

4 **Put the steps for cracking a long Caesar message by counting letters in order.**

☐ Shift every letter back and check it reads

☐ Work out the shift from it to E (or alif)

☐ Find the most common letter

☐ Count how often each letter appears

5 **An attacker with a stolen password file can guess about ten billion times a second. About how long would it take to try every password made of eight random lower-case letters?**

◯ About 20 seconds ◯ About 20 days ◯ About 90 years